University of BRISTOL

Loop shaping: lead and lag

Lecture 4 · CADE30008 Flight Dynamics & Advanced Control

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University of BRISTOL

From your design logs

Ten minutes, before we start

What came back

- **Where people are.** Most of you have A1.1 and A1.2 under way, several have the strip theory done, and a good number have been in M.003 with the simulators. A few have read the NASA reports end to end.
- **What is worrying people.** How much detail is wanted, whether the Quanser data is good enough to use, and where the aircraft data comes from.

Clarifications

- The brief is open on purpose. There is no single expected answer, and the rubric rewards the reasoning rather than the result.
- Imperfect data is fine, as long as you say so and say why. This demonstrates your critique, and B2.2 asks for it directly.

Your questions

Does my aircraft have to be one the simulator has?	**No.** A2.1 and A2.2 need simulator data, so the aircraft you *fly* has to be one X-Plane has. A1.2 is free, and you may change aircraft or model between questions
Where do I get mass, thrust, wing data?	NASA CR-2144, *Aircraft handling qualities data*, is **one option** among many. Search broadly, and evaluate what you find against what you need
Can I start Section B before A2.3 is done?	**Yes, and start now.** B1.1 and B1.2 are 15 marks between them and need no model at all. Only B2.1 waits for A2.3
How much Quanser detail?	An example of the design cycle, not a lab report. B1.1 says your own data or the example data
My 2% settling time looks wrong	On a 7.5° step, 2% is 0.15°. Consider the context: hinge friction, encoder resolution, and the draughts in M.003 off the other rigs and the ventilation

Whatever you choose, tell us what you are doing, in Section A and again in Section B. The two are marked separately, so each has to stand on its own.

Sources you found

Between you, these came up. All four are free from NASA's technical reports server.

- **Aircraft handling qualities data** (CR-2144) — mass, geometry and derivatives for a range of transports. **One option, not the answer**: find others and judge which suits your aircraft
- **Development and flight test of an emergency flight control system using only engine thrust** — the MD-11, and reference [4] in your brief
- **Flight test of a propulsion-based emergency control system on the MD-11** — the landings
- **The use of pilot rating in the evaluation of aircraft handling qualities** — Cooper and Harper's own paper, which A1.1 asks you to refer to

Also found: the NTSB report on United 232 and an ISASI paper on the Baghdad A300.

Not an exhaustive list. It is what turned up between you in one week, and there are leads in it worth following.

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Loop shaping: lead and lag

One more transfer function in the loop

MOTIVATIONS

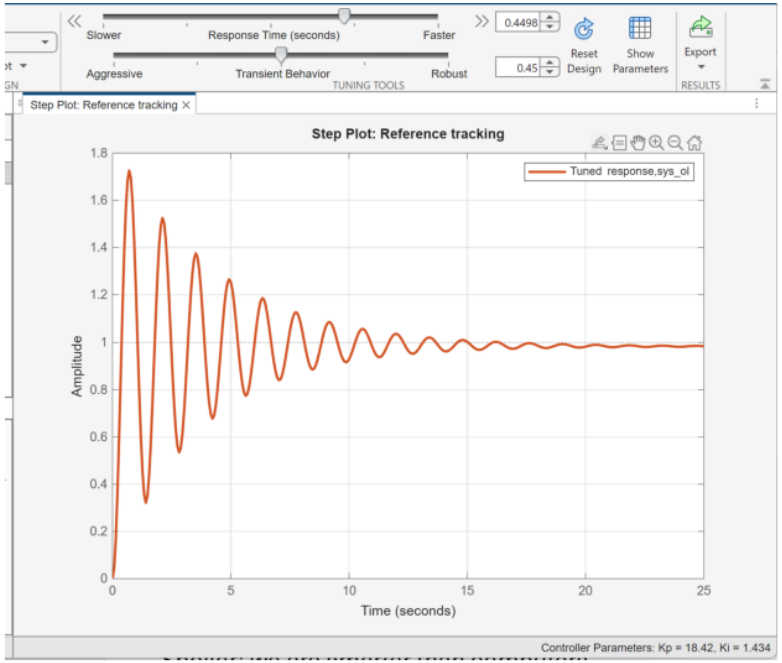
- First example of single-input, single-output design in the frequency domain.
- Illustrate how control design for ‘full fidelity’ pitch dynamics works.

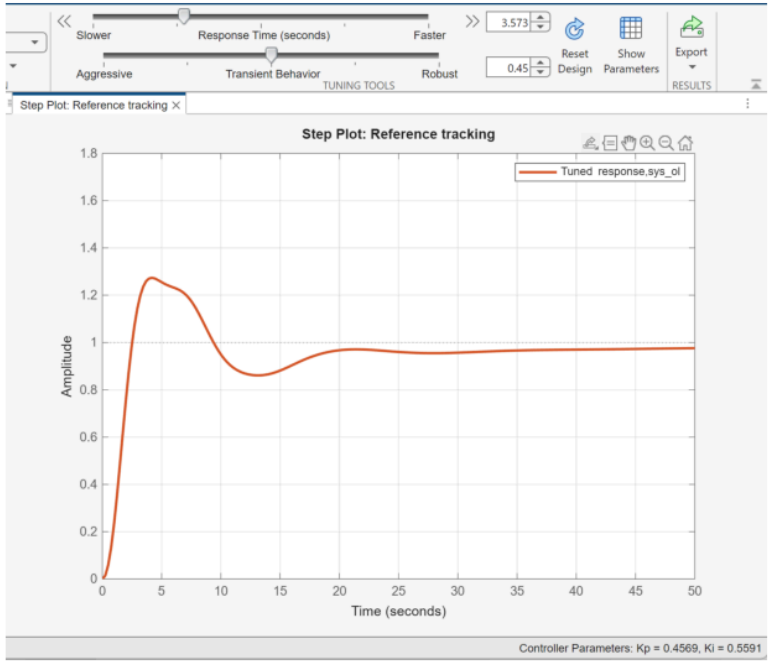
RECAP

Pitch dynamics

$$\frac{\theta(s)}{\delta_e(s)} = \frac{(s + 0.5877)\,(s + 0.02662)}{(s^2 + 1.127\,s + 1.862)(s^2 + 0.01579\,s + 0.01475)} \left(\frac{deg}{deg}\right)$$

Closed-loop response unsatisfactory using PID alone.





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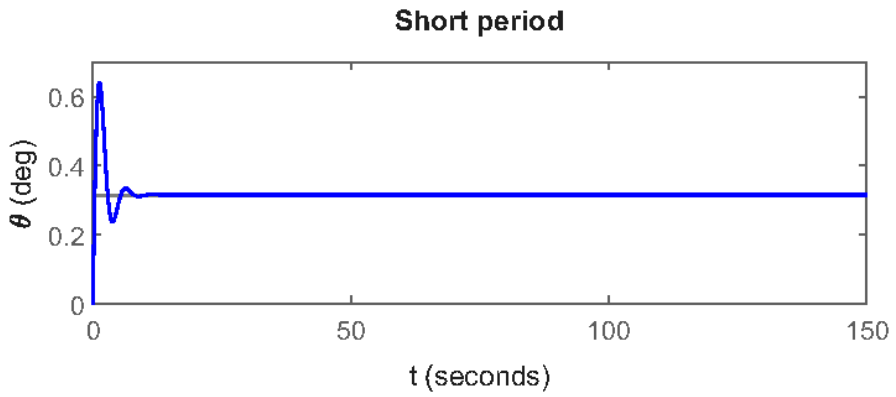
RECAP

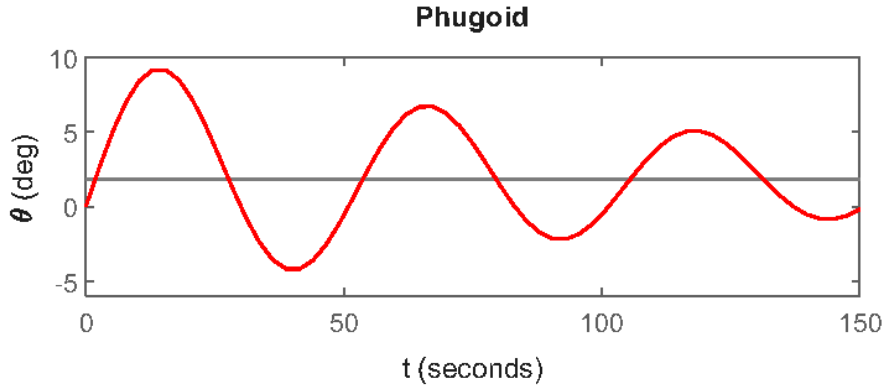
$$\frac{\theta(s)}{\delta_e(s)} = \underbrace{\frac{s + 0.5877}{s^2 + 1.127\,s + 1.862}}_{\text{short period}} \times \underbrace{\frac{s + 0.02662}{s^2 + 0.01579\,s + 0.01475}}_{\text{phugoid}}$$

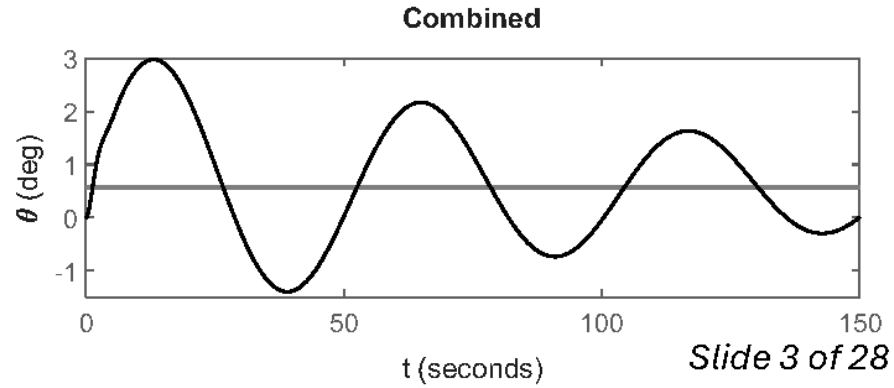
Fourth order dynamics with two modes spanning two different time frames.

Add a PI controller

$$C(s) = 2.2 + \frac{8.3}{s}$$







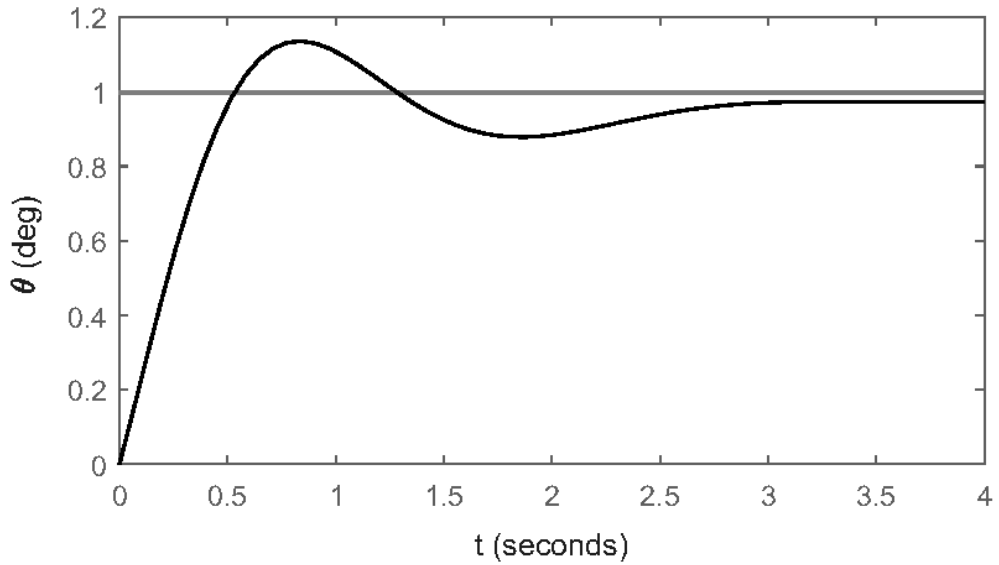
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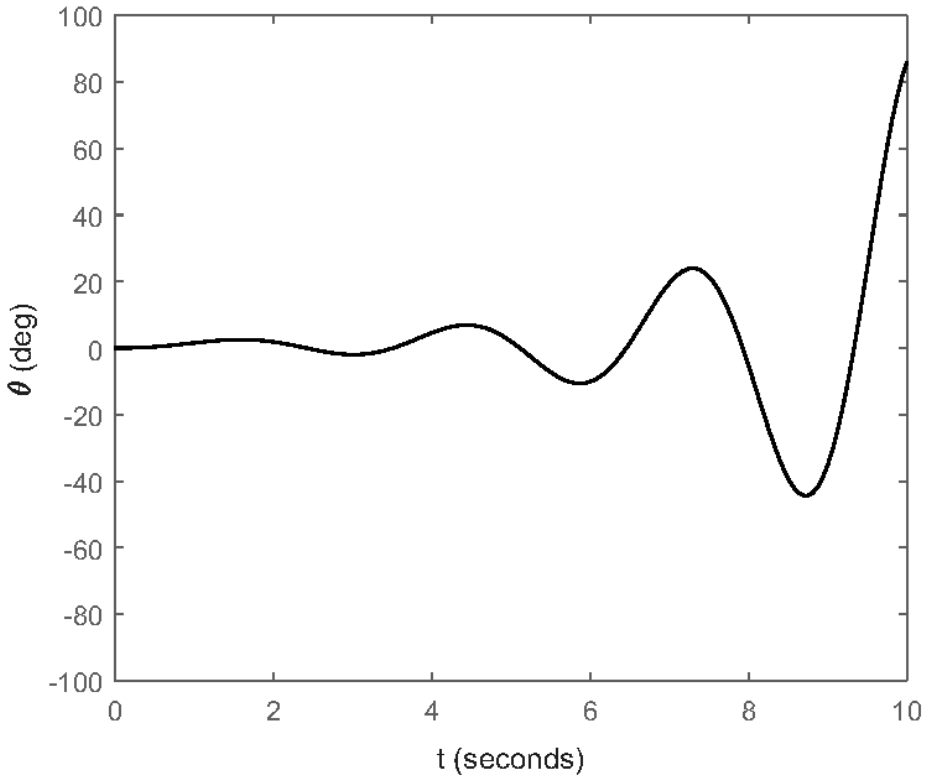
RECAP

Closed-loop response: plant + controller

$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127\,s + 1.862}\right)$$

$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127\,s + 1.862}\right)\left(\frac{s + 0.02662}{s^2 + 0.01579\,s + 0.01475}\right)$$

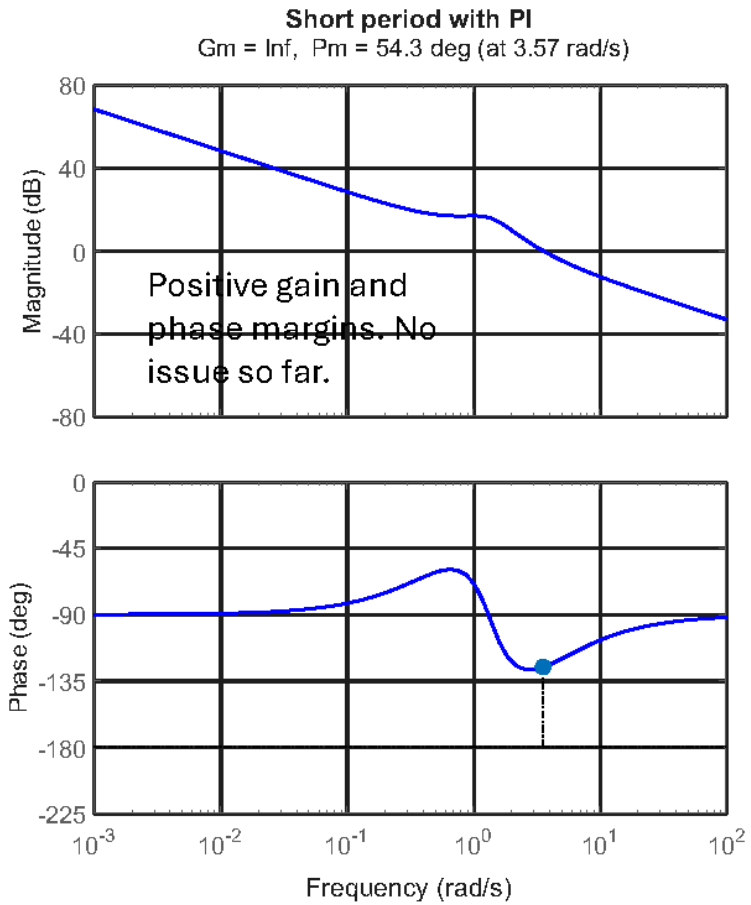




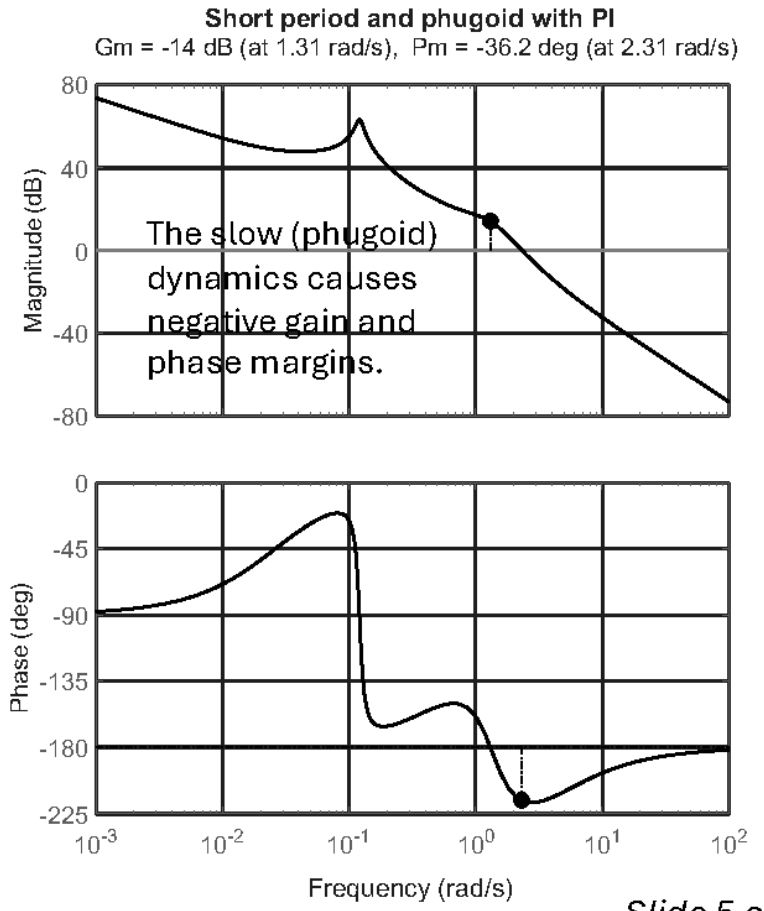
RECAP

Frequency response: plant + controller

$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127\,s + 1.862}\right)$$



$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127\,s + 1.862}\right)\left(\frac{s + 0.02662}{s^2 + 0.01579\,s + 0.01475}\right)$$



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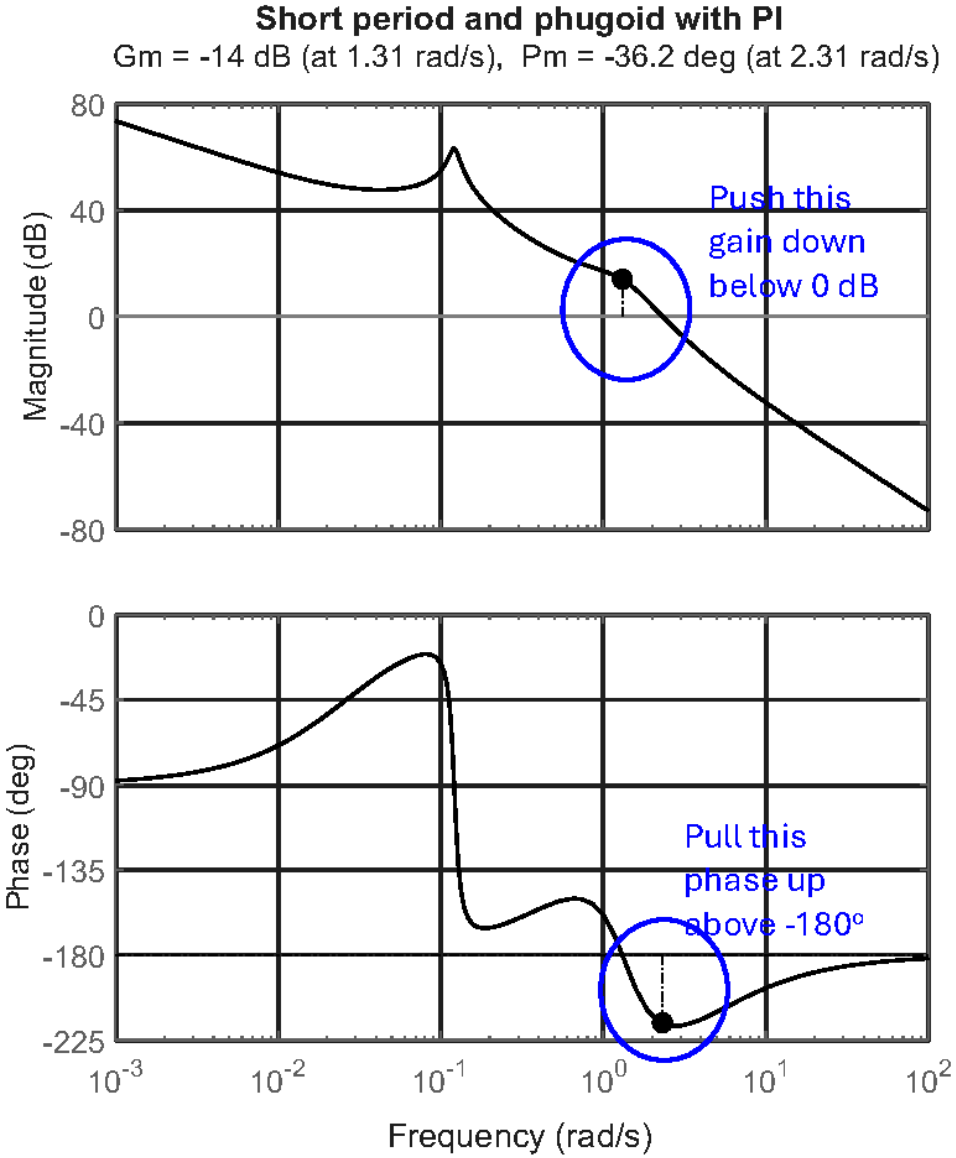
LOOP SHAPING

Add an additional transfer function to manipulate the frequency response.

$$L(s)C(s)\frac{\theta(s)}{\delta_e(s)}$$

$L(s)$: lead/lag compensator

Task: design $L(s)$ to raise the gain and phase margins to positive values.



LOOP SHAPING

General form of a lead/lag compensator

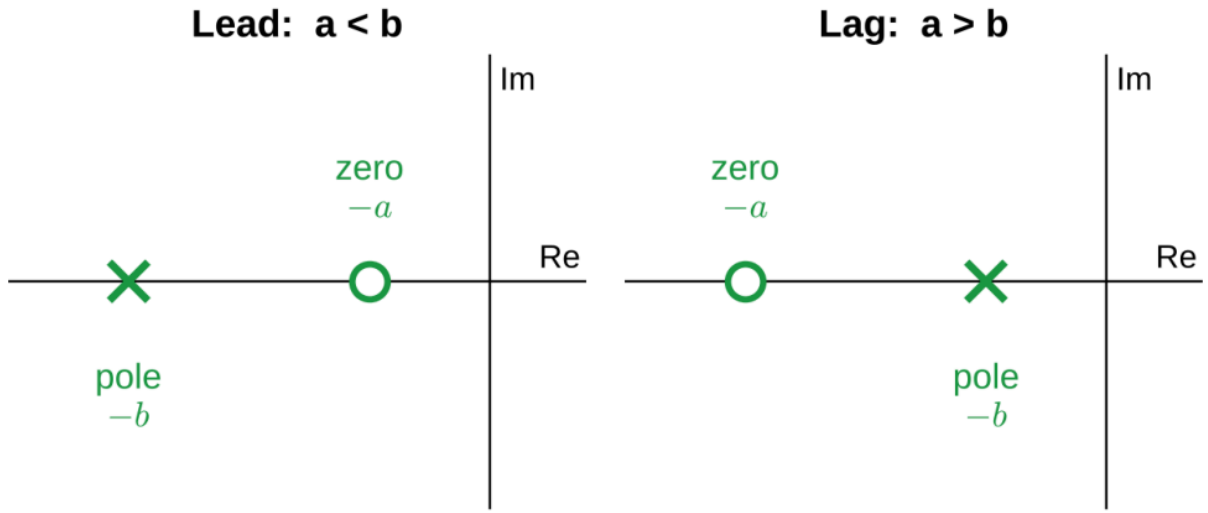
$$L(s) = \frac{\frac{s}{a} + 1}{\frac{s}{b} + 1} = \frac{b}{a}\frac{(s+a)}{(s+b)}$$

1 zero at s = −a

1 pole at s = −b

Lead compensator: a < b

Lag compensator a > b



Lead: zero nearer the origin. Lag: pole nearer the origin.

Final value theorem: set s = 0

$$L(0) = 1$$

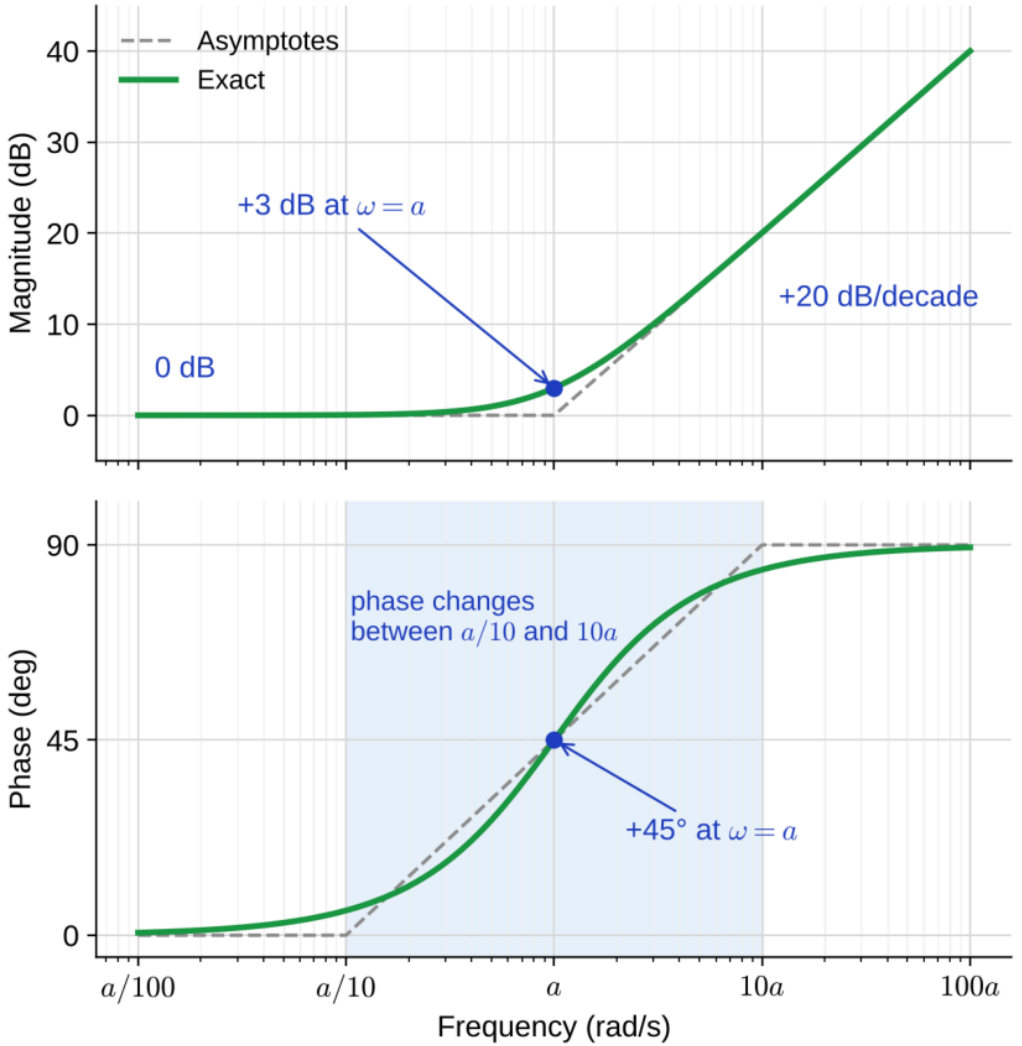
The compensator does not influence the steady-state value of the system it controls. Only the transient response is affected.

BUILDING BLOCK 1: ZERO

$$L(s) = \frac{\frac{s}{a} + 1}{\frac{s}{b} + 1}$$

Increase the gain starting from frequency a rad/s.

Add 90° in phase at high frequencies.

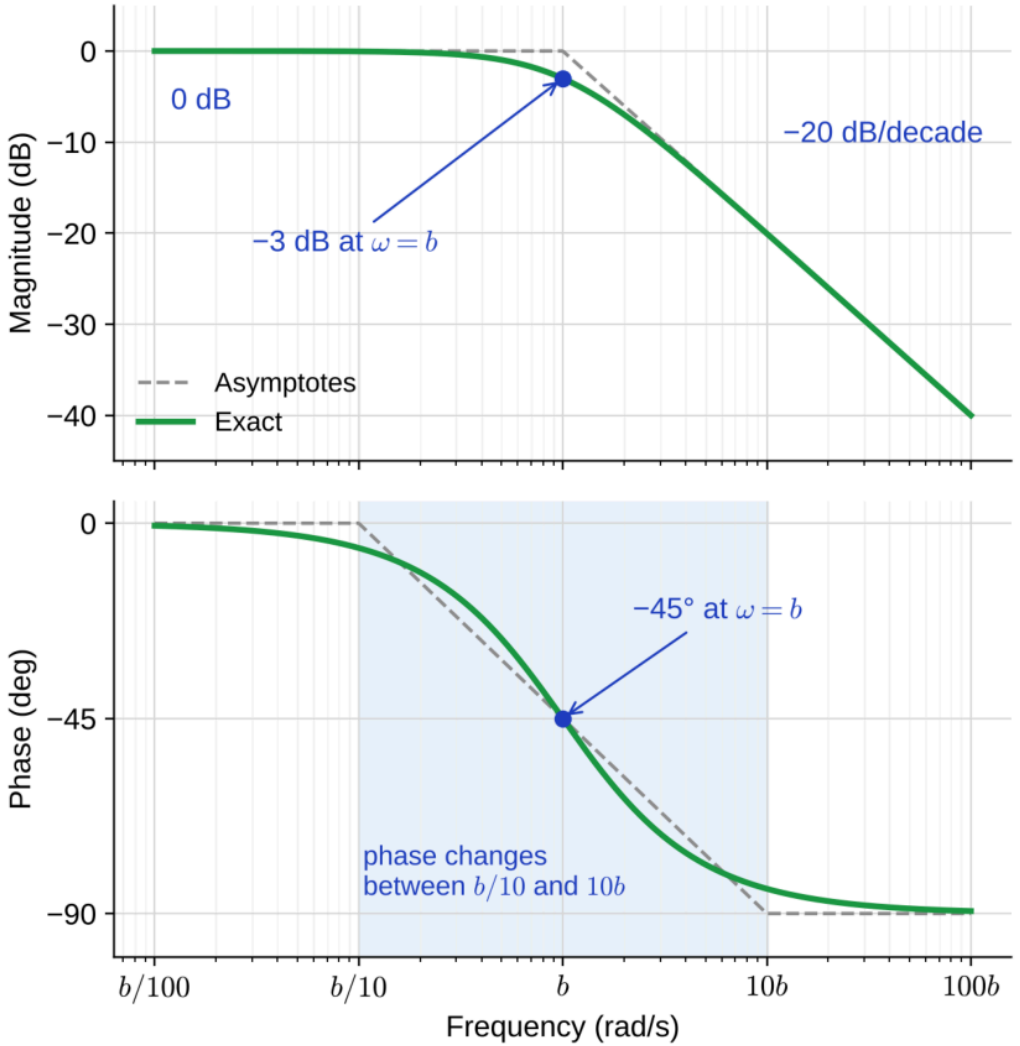


BUILDING BLOCK 2: POLE

$$L(s) = \frac{\frac{s}{a} + 1}{\frac{s}{b} + 1}$$

The opposite: reduce gain starting from frequency b rad/s.

Remove 90° in phase at high frequencies.

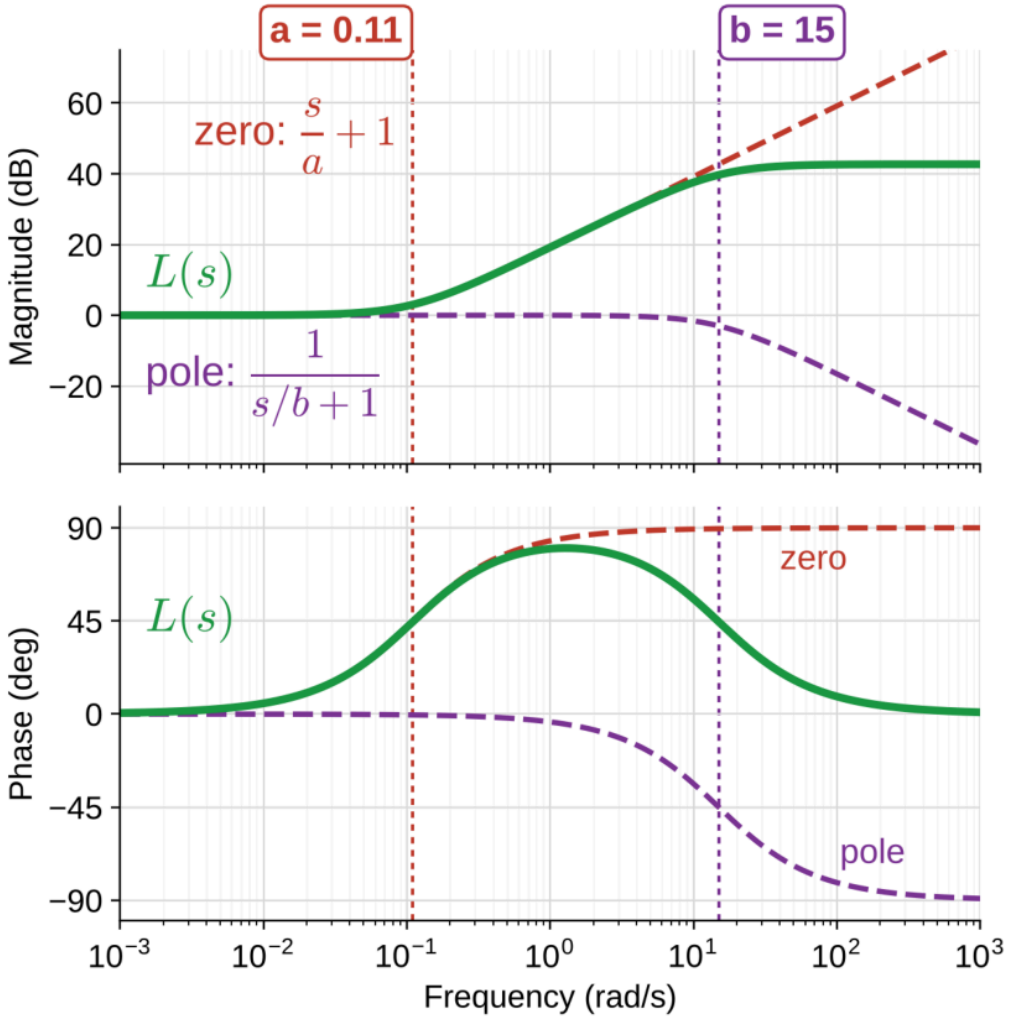


LEAD COMPENSATOR

$$L(s) = \frac{\frac{s}{a} + 1}{\frac{s}{b} + 1}$$

a < b by definition

Example: a = 0.11, b = 15

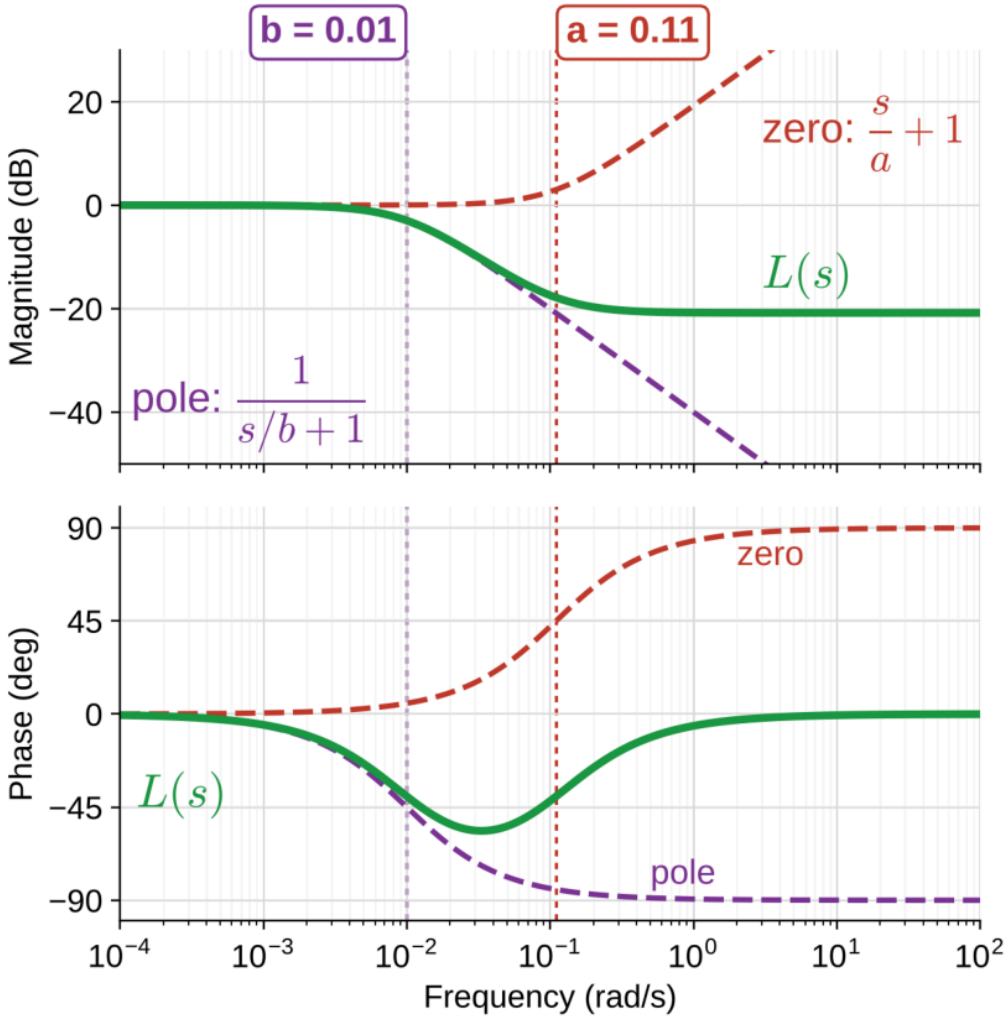


LAG COMPENSATOR

$$L(s) = \frac{\frac{s}{a} + 1}{\frac{s}{b} + 1}$$

a > b by definition

Example: a = 0.11, b = 0.01



COMBINED LEAD-LAG COMPENSATOR

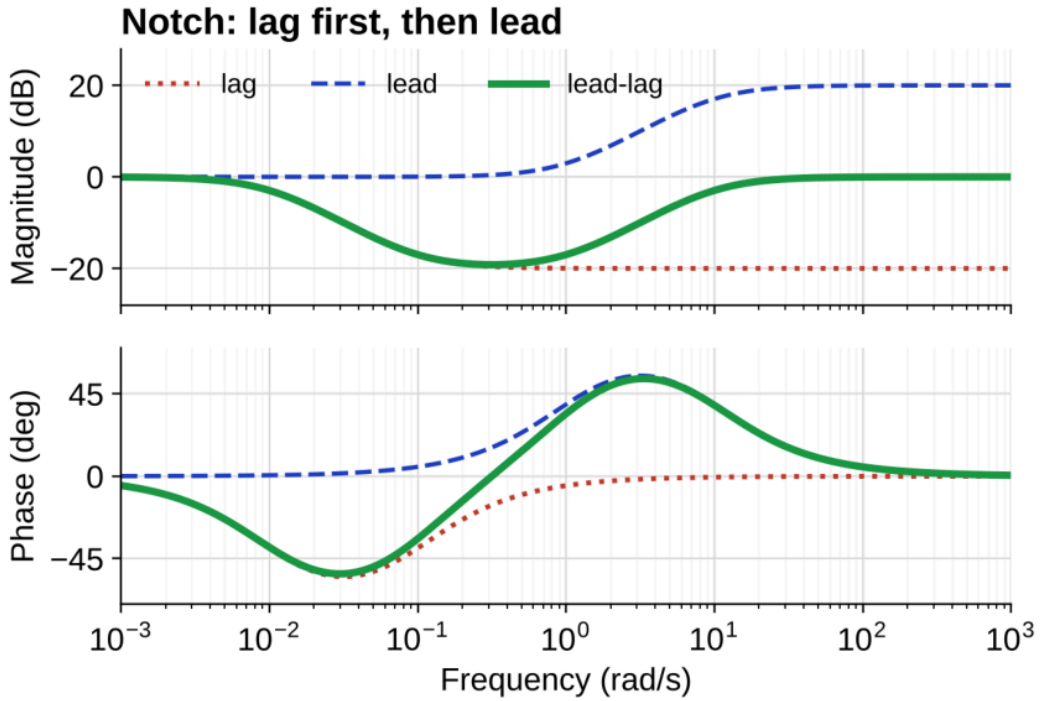
Put both components in series:

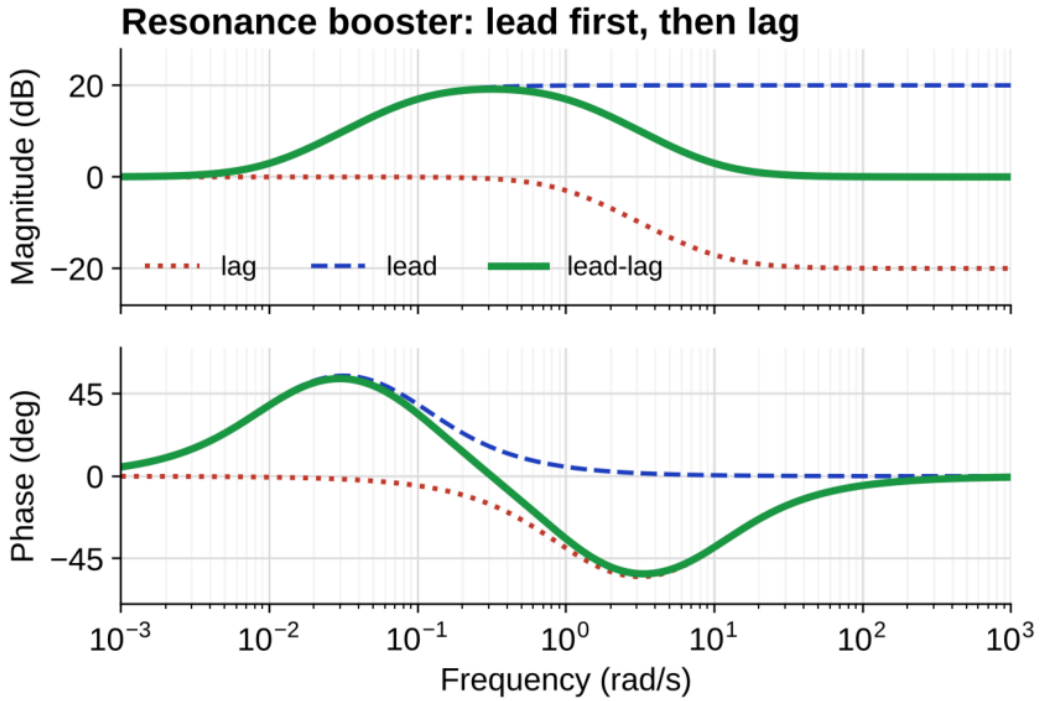
$$L(s) = L_{\text{lag}}(s)\, L_{\text{lead}}(s) = \left(\frac{\frac{s}{a_1}+1}{\frac{s}{b_1}+1}\right)\left(\frac{\frac{s}{a_2}+1}{\frac{s}{b_2}+1}\right)$$

The order of the two parts sets the shape:

- **Lag below lead:** notch filter.
- **Lead below lag:** a resonance booster.

Achieved by adjusting 4 variables: a_1, a_2, b_1, b_2

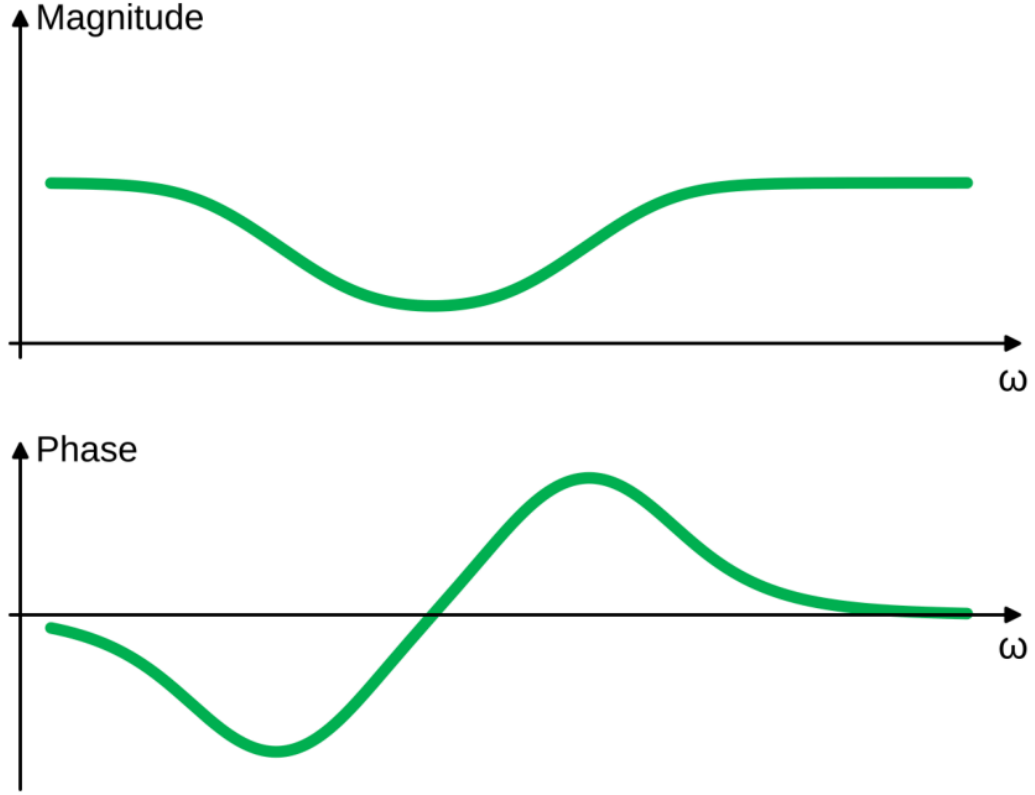




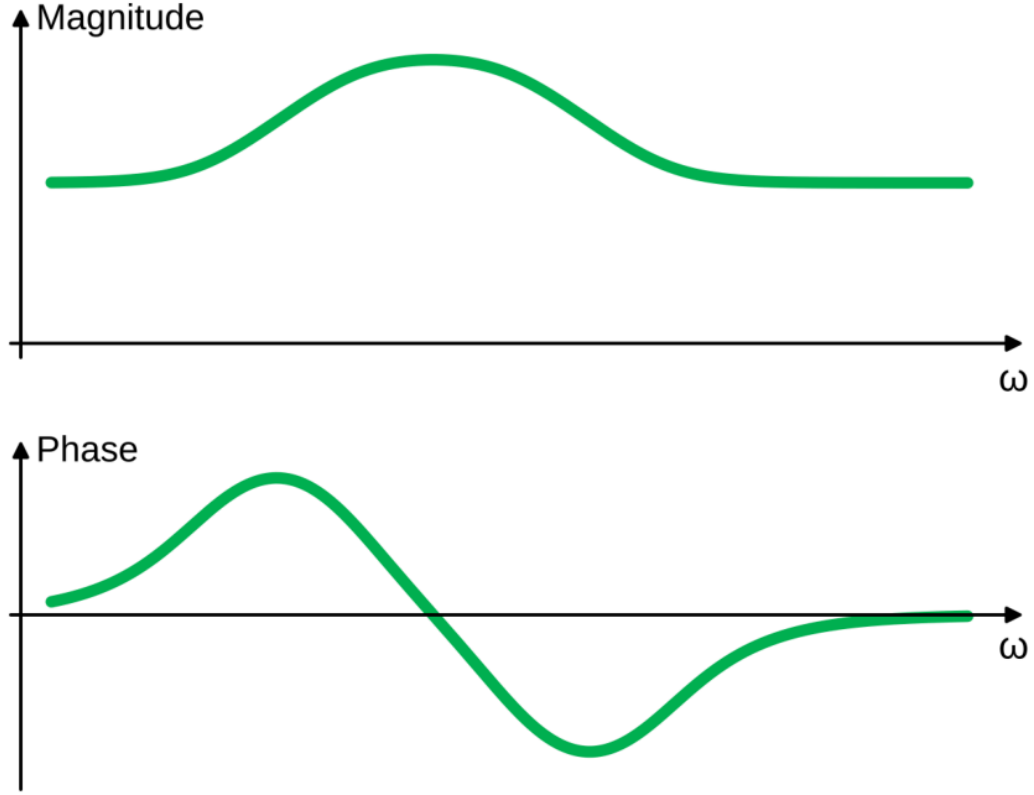
COMPENSATOR DESIGN

Two general shapes of a lead-lag compensator

Notch (lag → lead)



Resonance booster (lead → lag)

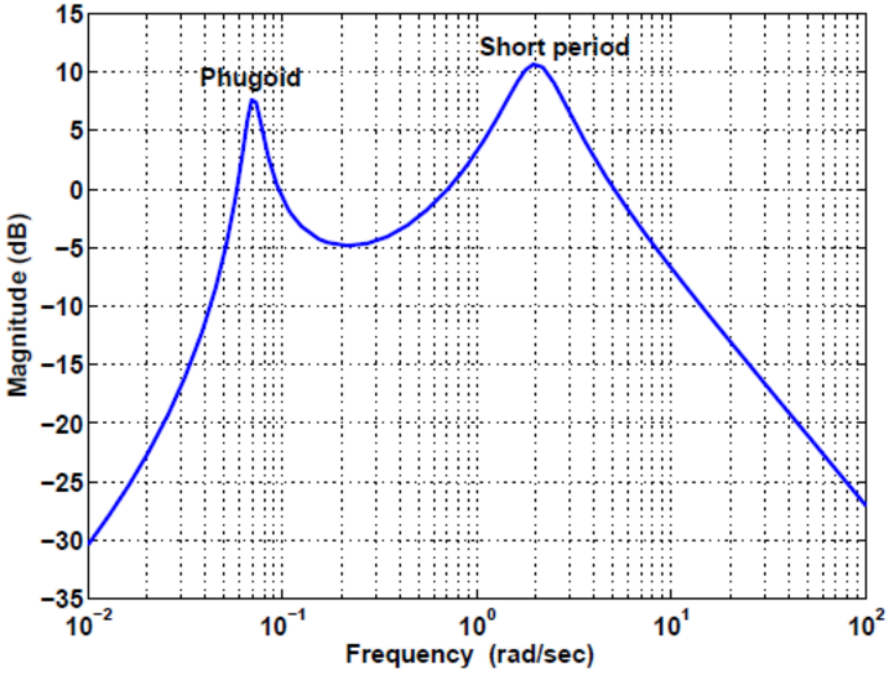


What can we do with them?

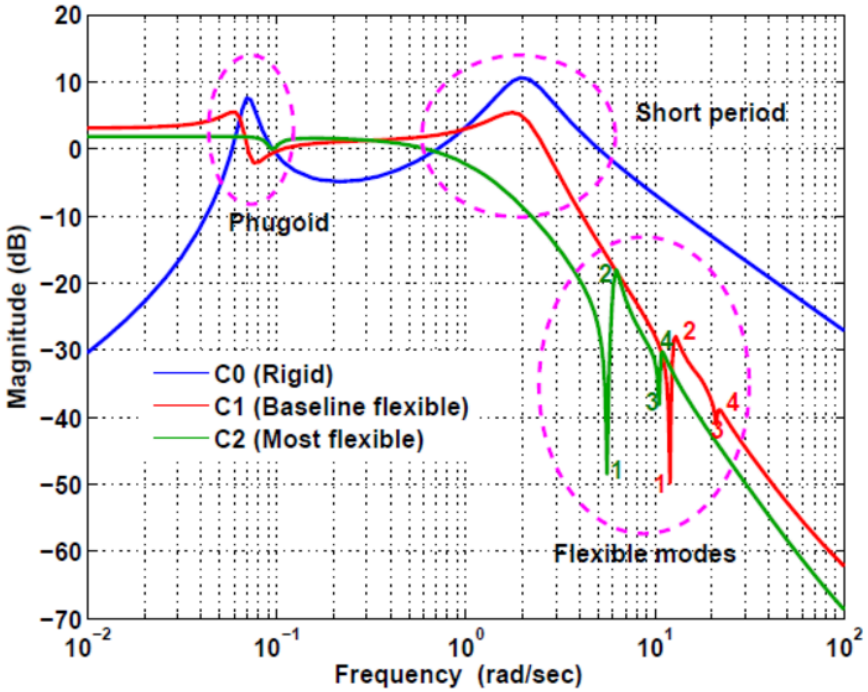
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USE 1: STRUCTURAL COUPLING

Usually mitigated by a notch filter



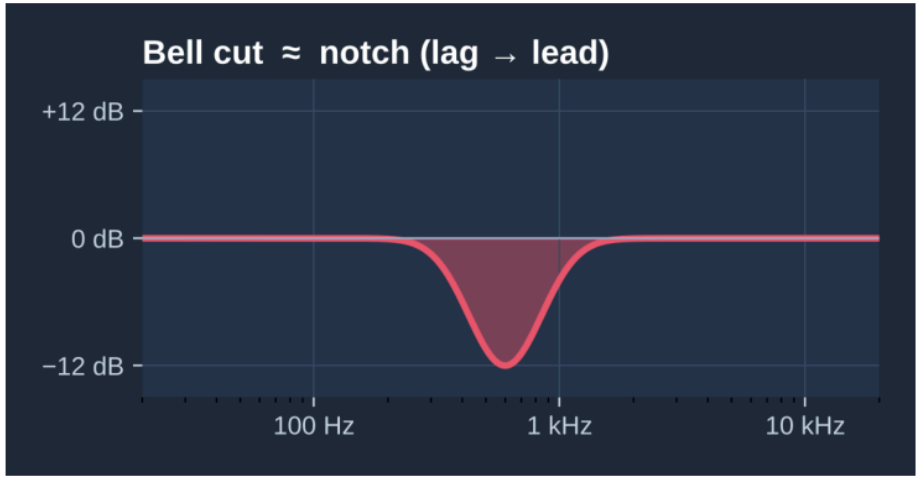
Rigid aircraft

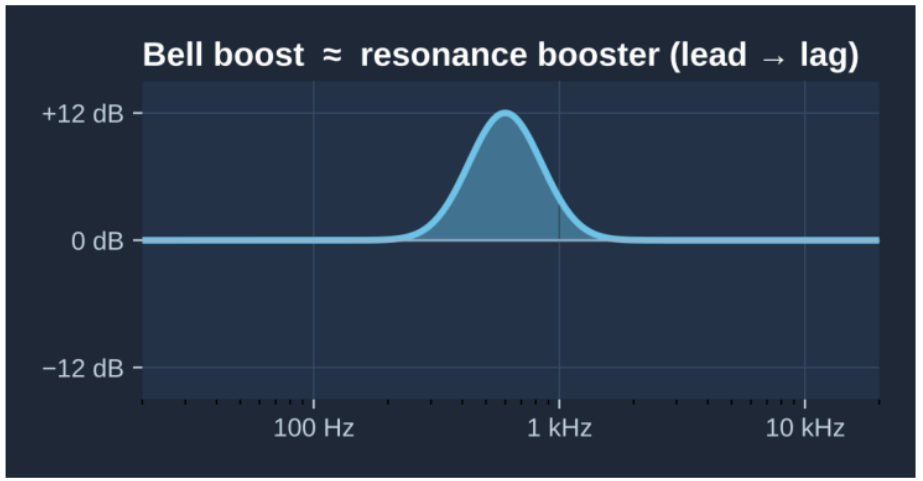


Flexible aircraft

USE 2: AUDIO PROCESSING







Equalisers (EQs) use poles and zeros:

- Low-pass filter: pole only (lag)
- High-pass filter: zero only (lead)
- Band pass/resonance boost: lead-lag compensator

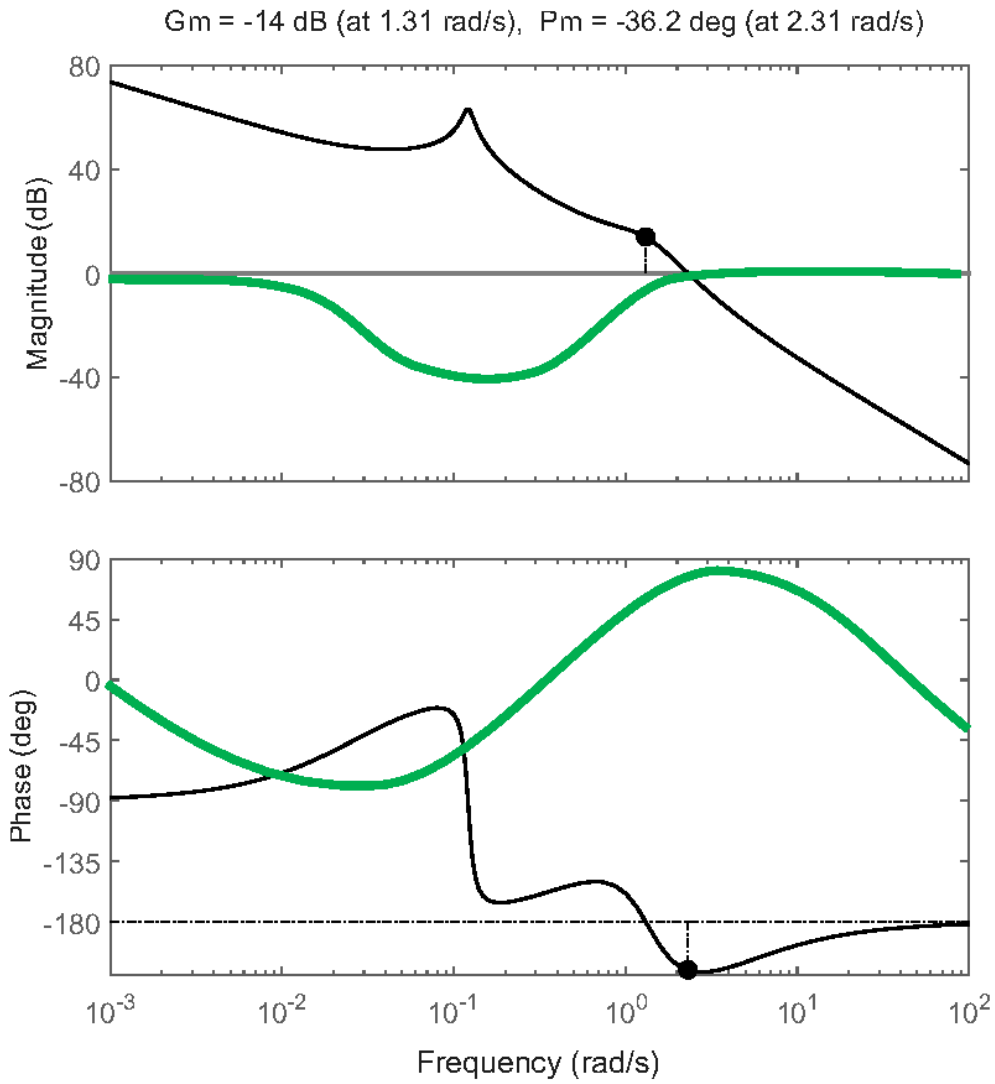
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FLIGHT CONTROL DESIGN

We want to attenuate the phugoid dynamics while keeping gain and phase margins positive.

Similar idea to the structural coupling problem above. Use a wide notch filter that targets two areas:

- Use the gain drop to lower the phugoid peak.
- Use the phase rise to raise the high-frequency phase lag above -180°.



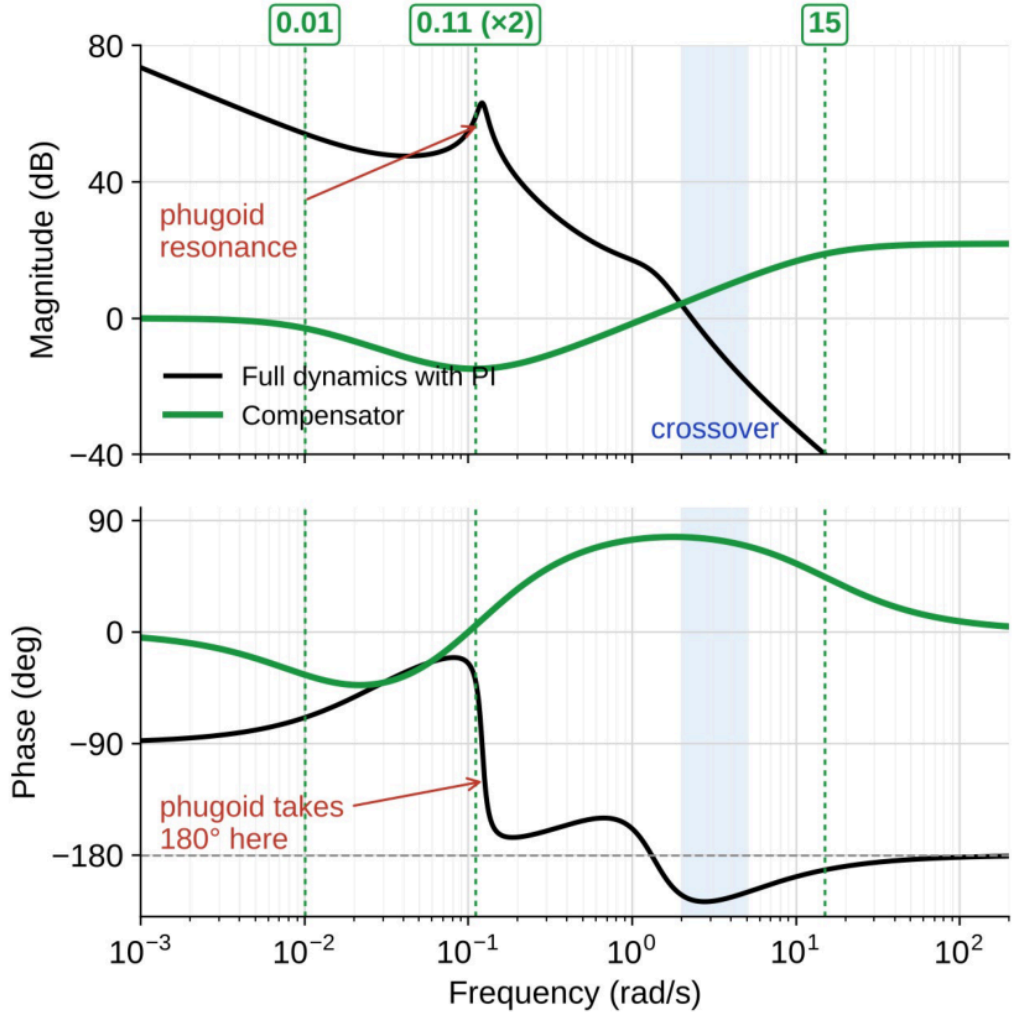
CHOOSING THE CORNER FREQUENCIES

$$L(s) = L_{\text{lag}}(s)\, L_{\text{lead}}(s) = \left(\frac{\frac{s}{a_1}+1}{\frac{s}{b_1}+1}\right)\left(\frac{\frac{s}{a_2}+1}{\frac{s}{b_2}+1}\right)$$

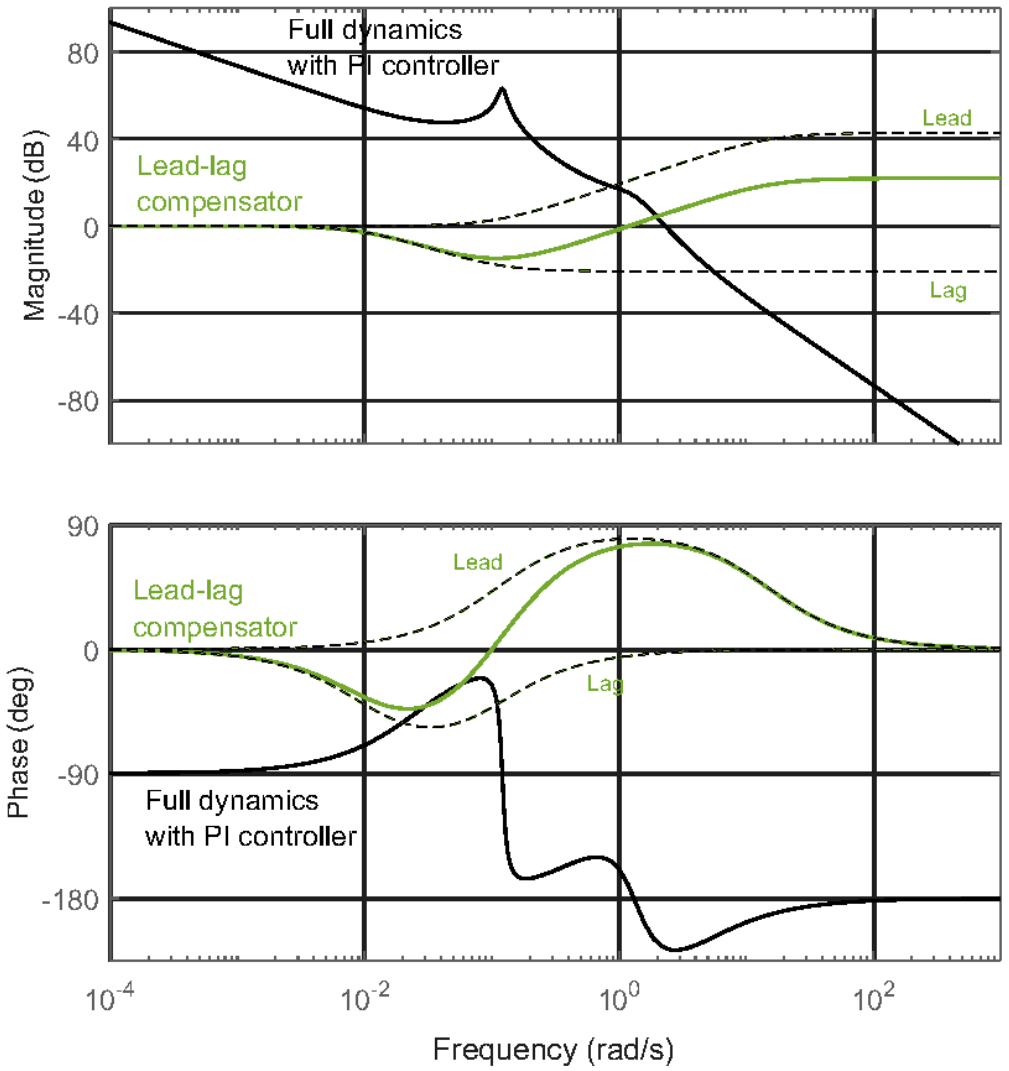
Adjust the 4 parameters based on the open-loop frequency response.

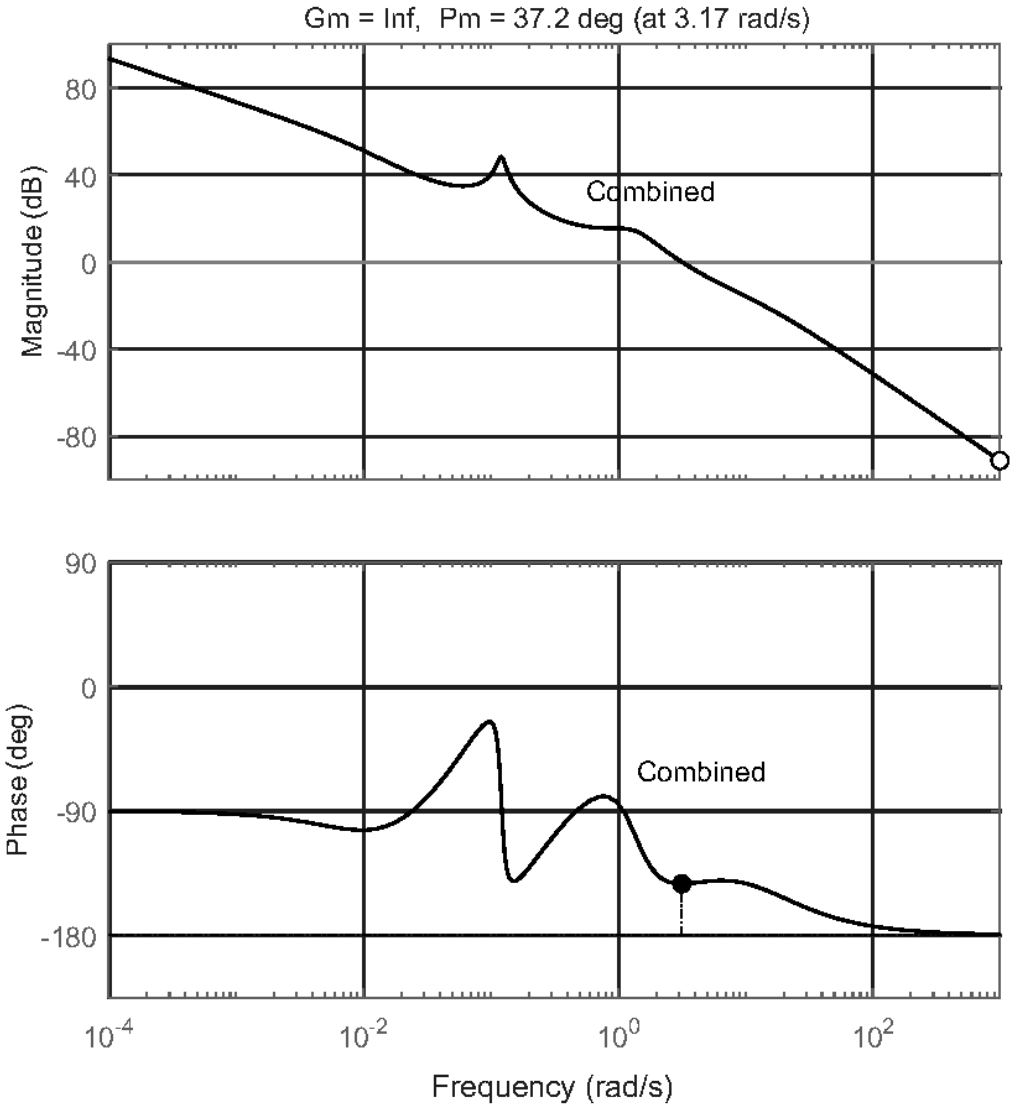
Final compensator

$$L(s) = \left(\frac{\frac{s}{0.11}+1}{\frac{s}{0.01}+1}\right)\left(\frac{\frac{s}{0.11}+1}{\frac{s}{15}+1}\right)$$



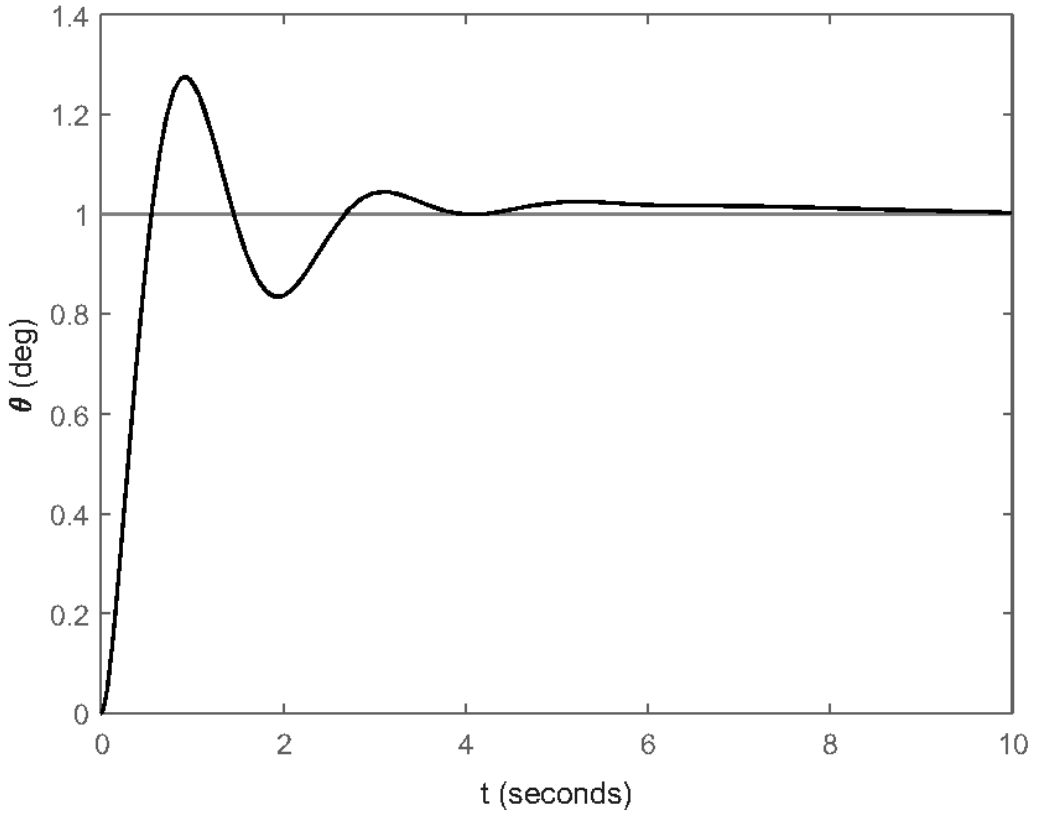
COMPENSATOR DESIGN

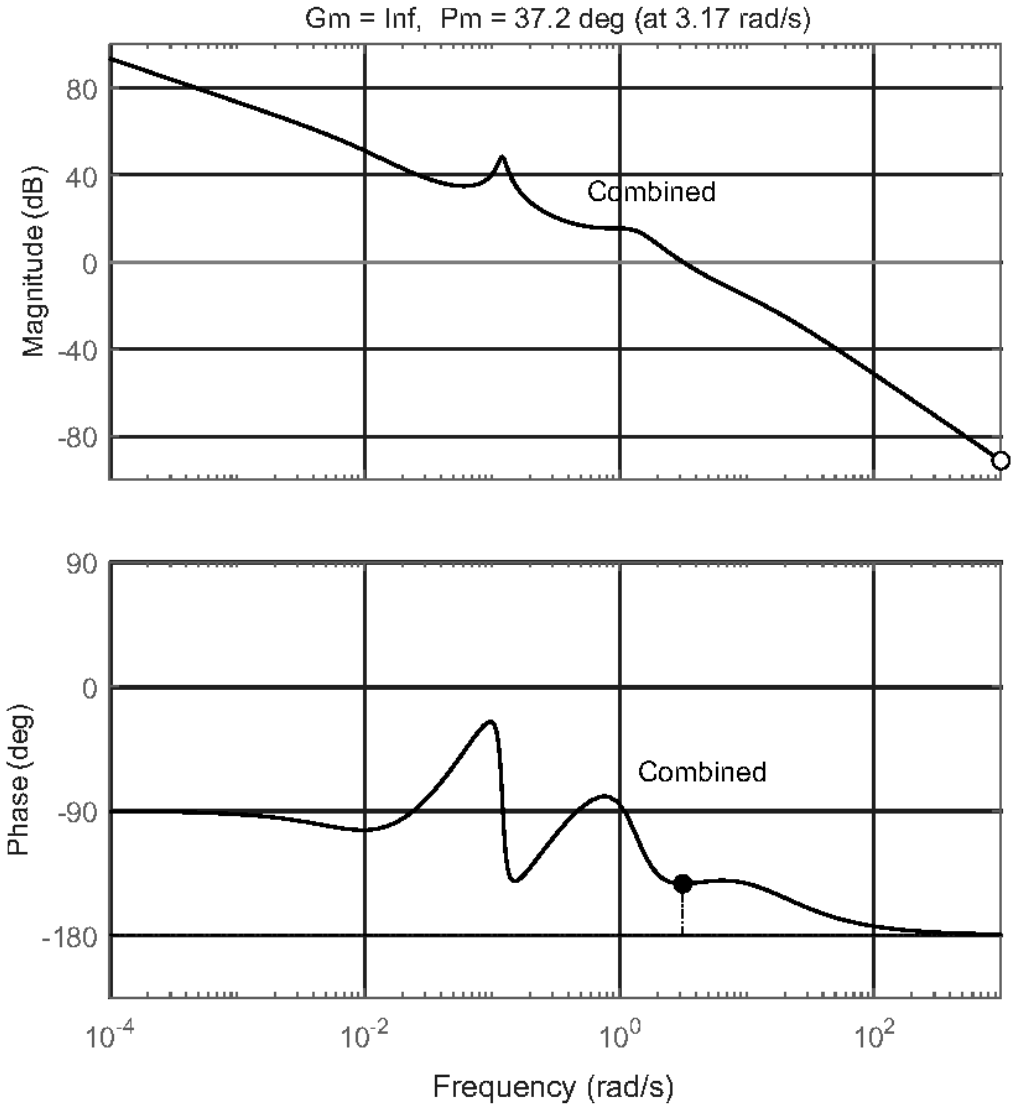




CLOSED-LOOP PERFORMANCE

- Full pitch dynamics
- PI controller designed for the short period mode only
- Compensator

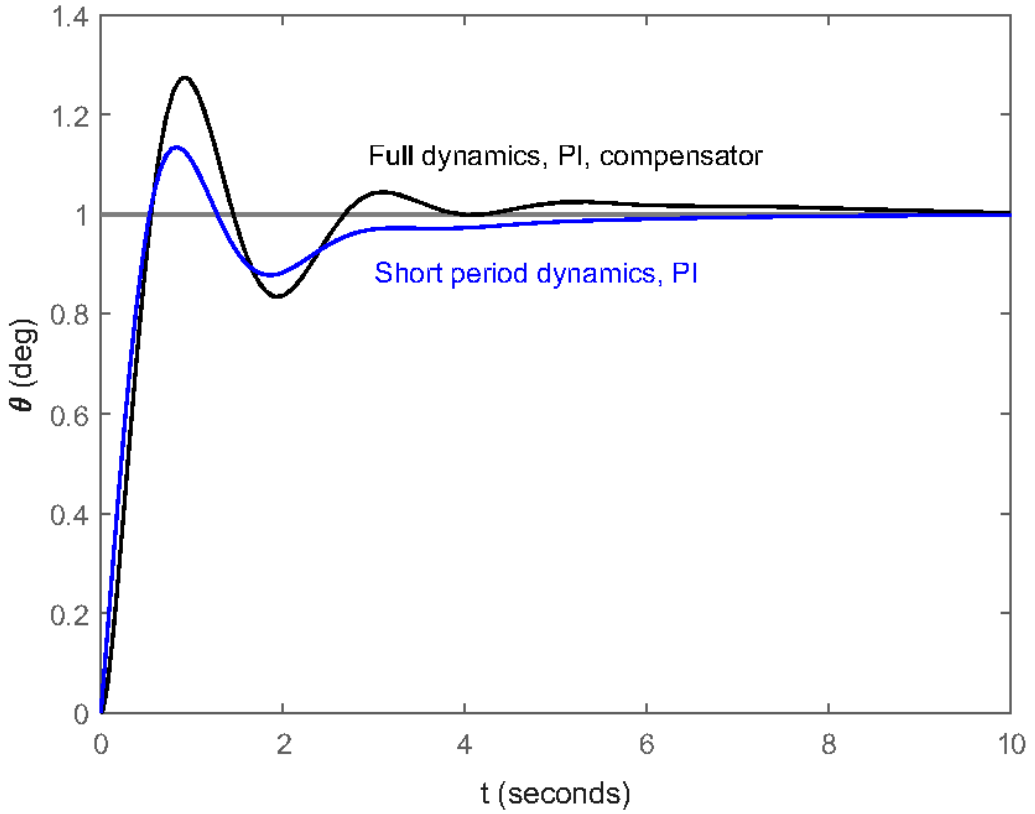


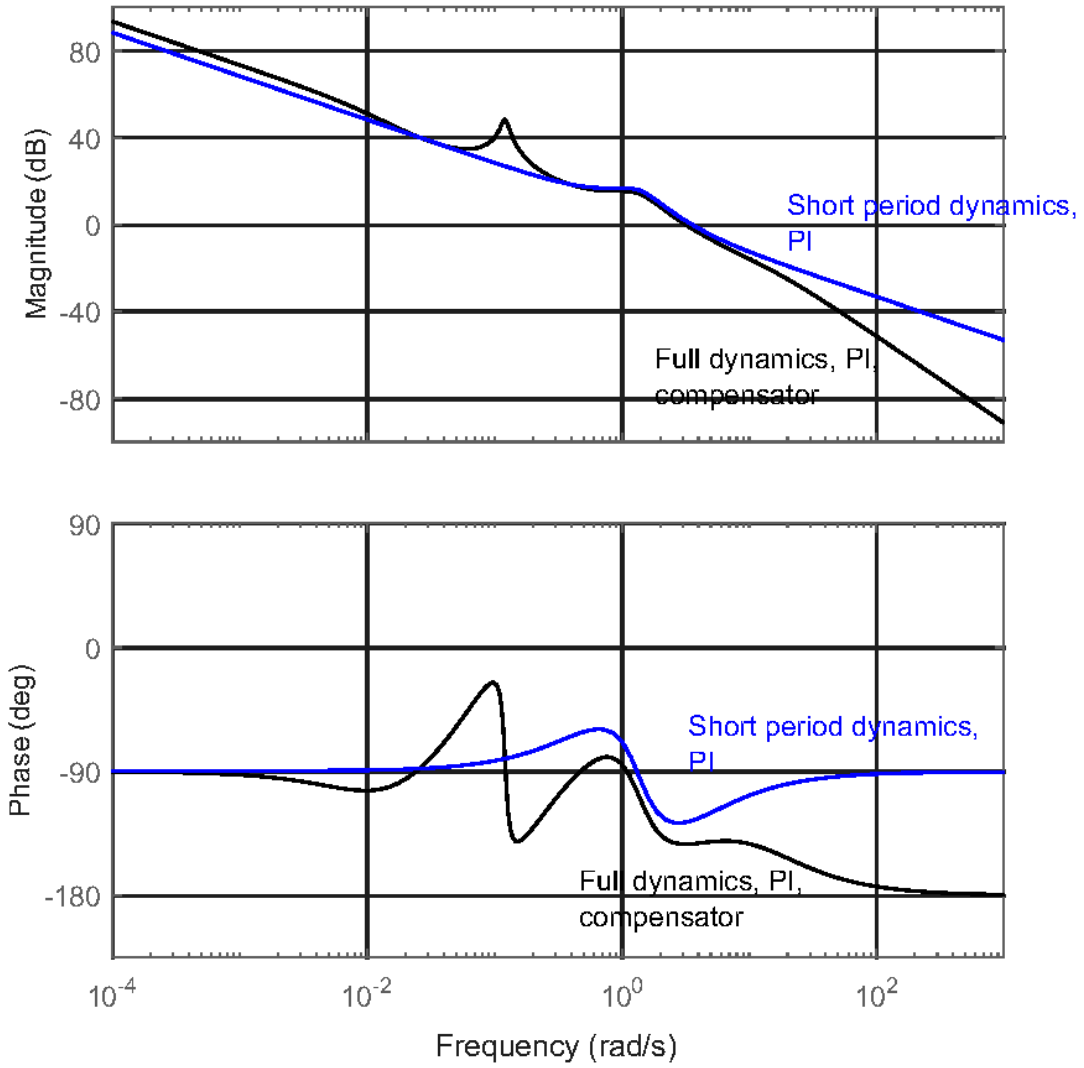


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CLOSED-LOOP PERFORMANCE

We've 'recovered' the performance previously seen on the reduced-order (short-period only) system.





AI-ASSISTED DESIGN

Level 1: design a controller for me. I don’t care (or don’t understand) what the open-loop dynamics is.

Level 2: I know what the controller should look like. Tune the parameters for better performance.

Level 3: give me the tool to design it myself.

AI-ASSISTED DESIGN

Level 1

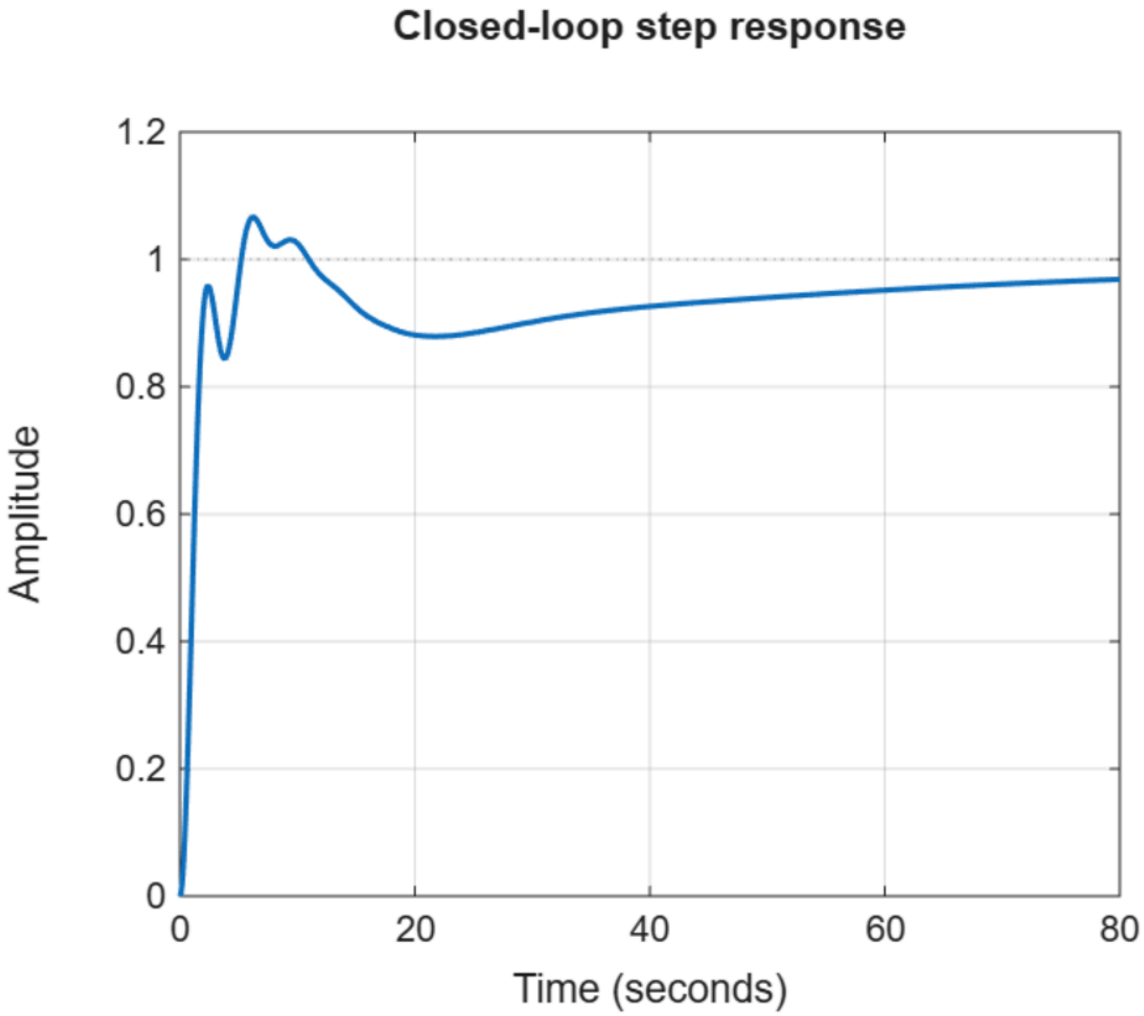
Prompt:

“part1 = (s+0.5877) / (s^2+1.127*s+1.862);
part2 = (s+0.02662) / (s^2+0.01579*s+0.01475);
ol_full = part1*part2;

That's my open-loop system. Design a feedback controller for me in matlab.”

Output: a PI with Kp = 1.2 and Ki = 0.2.

Too slow. We need to get to our destination within 5 seconds.



AI-ASSISTED DESIGN

Level 2

Prompt: “Your controller is too slow and shaky. We might need a lead/lag compensator to get rid of the low-frequency dynamics.”

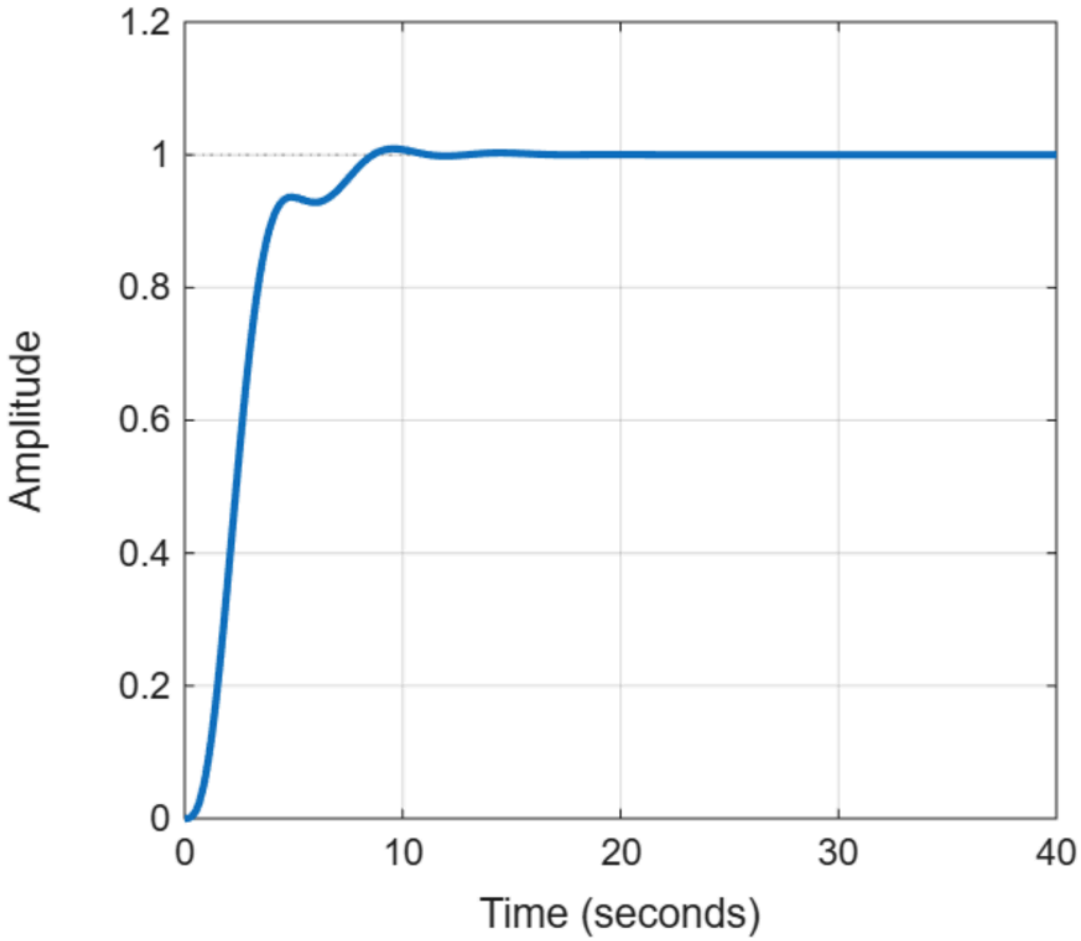
Output:

- A PI with Kp = 0 and Ki = 0.6
- A lead/lag compensator with transfer function

$$\frac{s^2 + 0.01579s + 0.01475}{(s + 0.5)\,(s + 0.02662)}$$

Issue 1: still too slow

Issue 2: does the compensator look familiar?



AI-ASSISTED DESIGN

Level 2

AI is cheating by algebraically cancelling out the phugoid dynamics.

Compensator:

$$\frac{s^2 + 0.01579s + 0.01475}{(s + 0.5)\,(s + 0.02662)}$$

Plant:

$$\frac{\theta(s)}{\delta_e(s)} = \frac{(s + 0.5877)\,(s + 0.02662)}{(s^2 + 1.127\,s + 1.862)(s^2 + 0.01579\,s + 0.01475)}$$

This is a very bad idea. Plant modelling is never perfect. This controller would break down in real life.

AI-ASSISTED DESIGN

Level 2

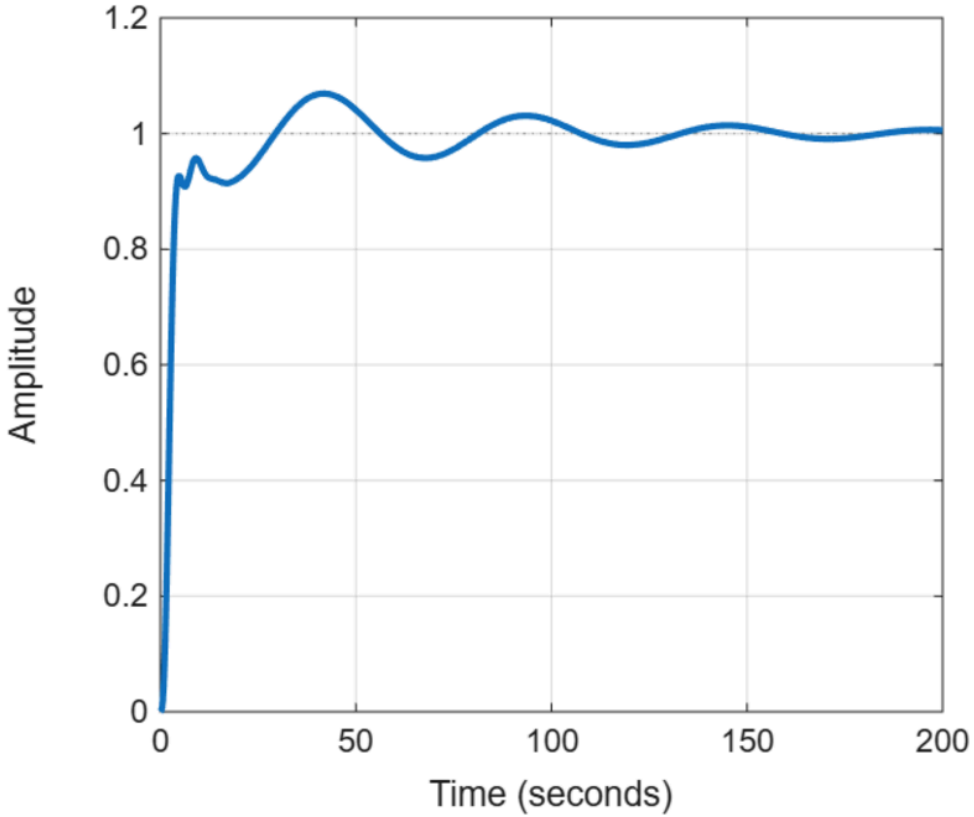
Quick test: slightly modify the plant

Plant:

$$\frac{\theta(s)}{\delta_e(s)} = \frac{(s + 0.5877)\ (s + 0.02662\ 0.03)}{(s^2 + 1.127\ s + 1.862)(s^2 + 0.01579\ s + 0.01475\ 0.02)}$$

Pole-zero cancellation misses its intended target.

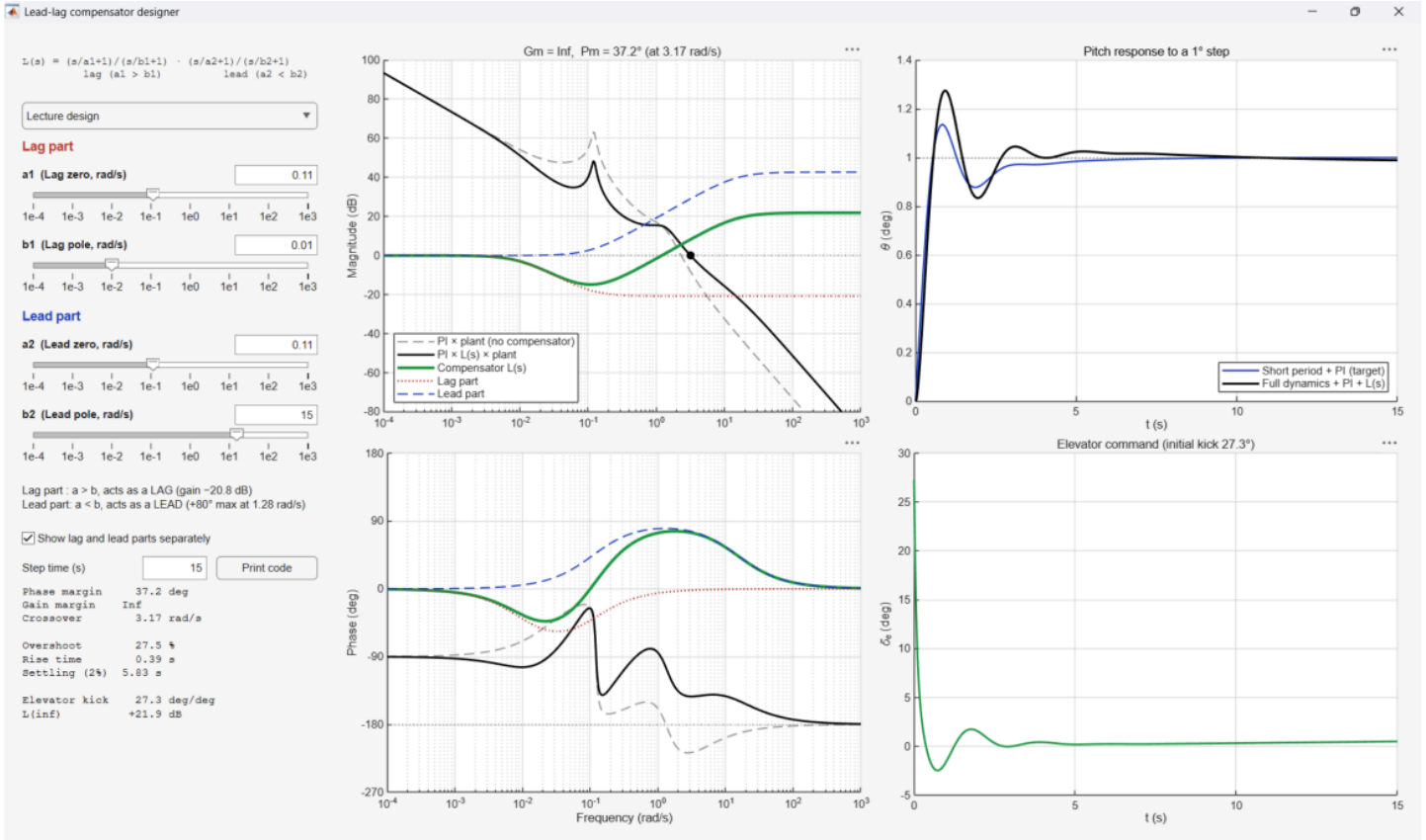
Closed-loop response with the modified plant unacceptable.



AI-ASSISTED DESIGN

Level 3

Prompt: “I’ve designed a PI controller for the short period dynamics only. The gains are Kp = 2.2 and Ki = 8.3. Give me an app to design the 4 parameters a1, b1, a2, b2 in a lead-lag compensator, to be added to the feedback loop. User will adjust the 4 variables. Show the frequency and step responses on the side.”



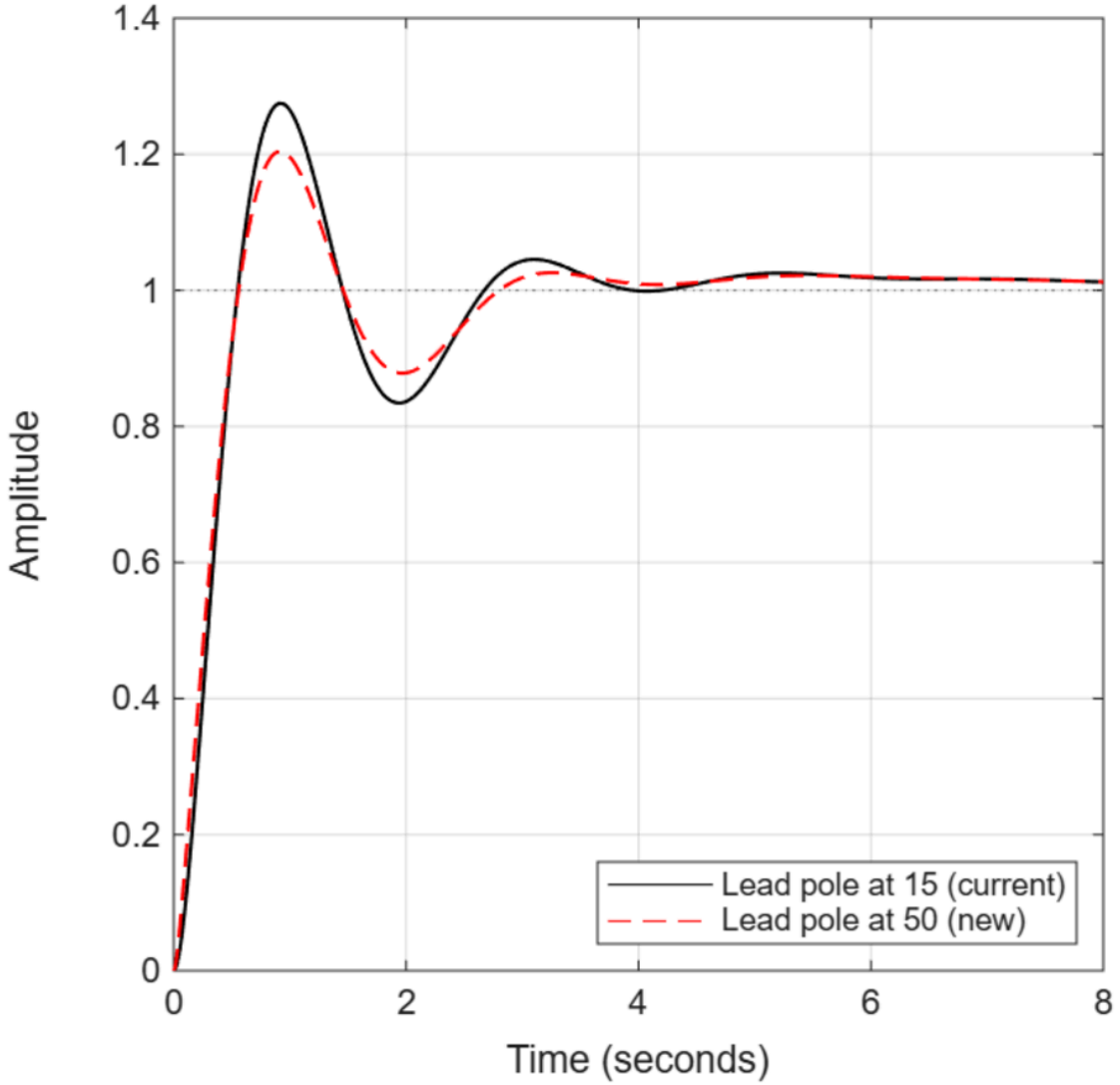
AI-ASSISTED DESIGN

Level 3

My pre-AI age controller:

$$C(s)L(s) = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{\frac{s}{0.11}+1}{\frac{s}{0.01}+1}\right)\left(\frac{\frac{s}{0.11}+1}{\frac{s}{15}+1}\right)$$

Changing the lead compensator's pole from 15 to 50 gives an even better response.



AI-ASSISTED DESIGN

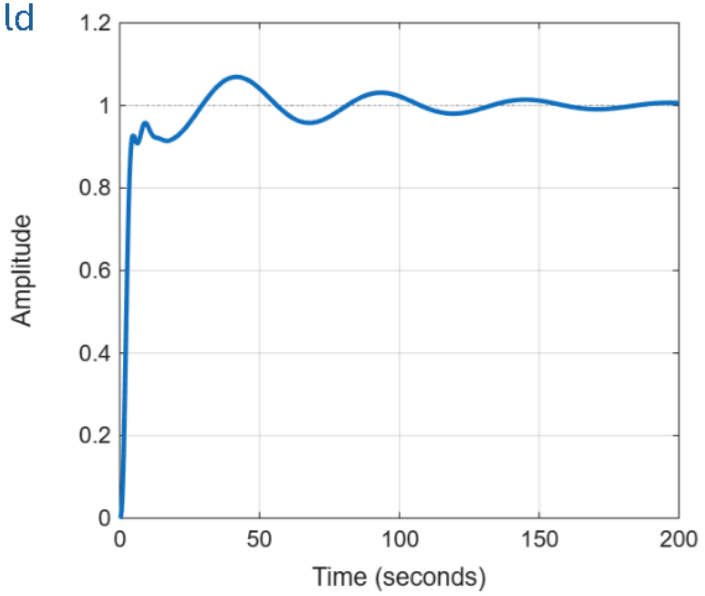
Level 3

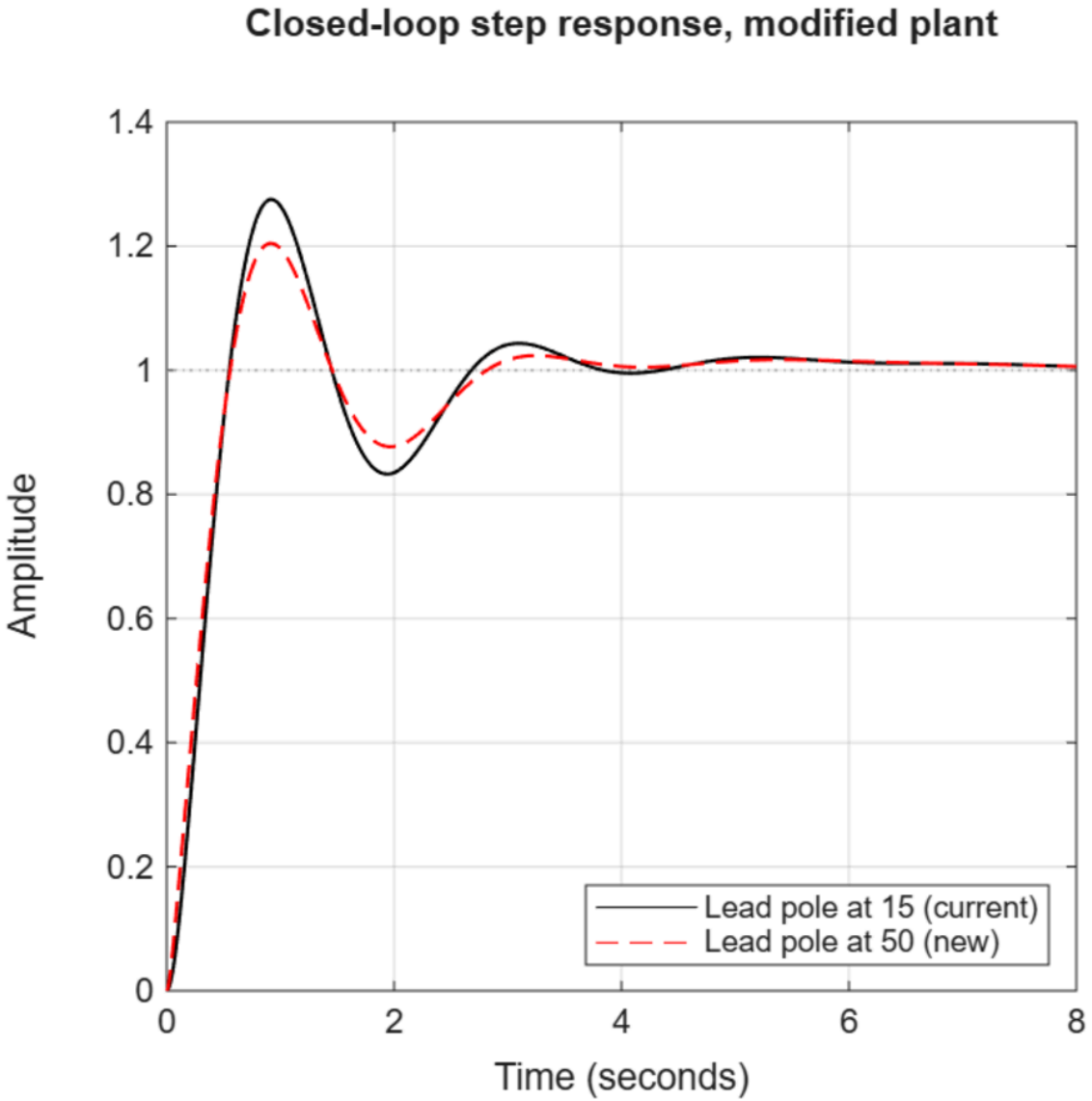
Test both the old and new controllers on the *modified* plant.

$$\frac{\theta(s)}{\delta_e(s)} = \frac{(s + 0.5877)\ (s + 0.02662\ 0.03)}{(s^2 + 1.127\ s + 1.862)(s^2 + 0.01579\ s + 0.01475\ 0.02)}$$

Barely any change in the closed-loop response.

Reminder: this is what would happen if you use the 'AI-powered' pole-zero-cancellation controller on the modified plant.





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Questions?